About Bootstrap

- Bootstrap is a free, open-source, and popular front-end framework for building responsive, mobile-first websites (for mobile devices – short screens) and web applications, providing pre-designed HTML, CSS, and JavaScript components for faster and easier web development.
- Responsive web design is about creating web sites which automatically adjust themselves to look good on all devices, from small phones to large desktops.
- Bootstrap comes with the Bootstrap classes, which means you don't have to create and design the components from scratch; instead, you can use the classes provided by Bootstrap and build on top of those classes.

How to get Bootstrap:

There are two ways to start using Bootstrap 4 on your own web site.

1. Include Bootstrap from a CDN
2. Download Bootstrap from getbootstrap.com

Bootstrap CDN

- If you don't want to download and host Bootstrap yourself, you can include it from a CDN (Content Delivery Network). **jsDelivr** provides CDN support for Bootstrap's CSS and JavaScript.

```
<!-- Bootstrap CSS CDN -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">

<!-- Bootstrap JS -->
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js">
</script>
```

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Ex: Develop a sample page in Django by integrating Bootstrap.

Step 1: Create a Project and Application

```
python -m django startproject BS_Project
cd BS_Project
python .\manage.py startapp SimpleBSApp
```

Step 2: Register the App

```
INSTALLED_APPS = [
                    ....
                    'SimpleBSApp',
]
```

Step 3: Create View in views.py file

```
def home(request):
        return render(request, 'home.html')
```

Step 4: Create URLs

```
from SimpleBSApp import views
urlpatterns = [
        path('admin/', admin.site.urls),
        path('',views.home)
]
```

Step 5: Create templates folder in main project folder. Create home.html file

Step 6: Add templates folder path ('F:\S1-S2\BS_Project\\templates') to 'DIRS' in 'TEMPLATES' variable.

Step 7: Add Bootstrap (Very Important Step) to home.html file (Code I taken from https://www.w3schools.com/bootstrap/bootstrap_buttons.asp)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
  <script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/js/bootstrap.min.js"></script>
</head>
<body>

<div class="container">
  <h2>Button Styles</h2>
  <button type="button" class="btn">Basic</button>
  <button type="button" class="btn btn-default">Default</button>
  <button type="button" class="btn btn-primary">Primary</button>
  <button type="button" class="btn btn-success">Success</button>
```


Suresh Yadlapati

```
  <button type="button" class="btn btn-info">Info</button>
  <button type="button" class="btn btn-warning">Warning</button>
  <button type="button" class="btn btn-danger">Danger</button>
  <button type="button" class="btn btn-link">Link</button>
</div>

</body>
</html>
```

Step 8: Run the server - python manage.py runserver

Output:

Create an application with Tables, grids using Bootstrap in Django

- Follow the Same as the previous steps
- Add any Grid and Table code in **home.html** file (https://www.w3schools.com/bootstrap)

Create a Django App with Carousels feature.

- Follow the Same as the previous steps
- Add Carousel code (https://www.w3schools.com/bootstrap/bootstrap_carousel.asp) in **carousel.html** file. Place all html files in templates folder.

  Copy the code from w3scools.com

  Don't forget to add the following code:

  {% load static %}

  and

  <img src="{% static "images/bec.jpeg" %}" style="width:100%;">
- Place the image files in static/images folder
- Add the following in settings.py file

```
STATIC_URL = '/static/'
STATICFILES_DIRS = [
    BASE_DIR / "static",
```

```
]
```

- Create View and map urls to view

  views.py

```
def home2(request):
        return render(request, 'carousel.html')
```

  urls.py

```
urlpatterns = [
        path('admin/', admin.site.urls),
        path('basicBS',views.home),
        path('Grid_Table',views.home1),
        path('Carousel',views.home2),
]
```

- Run the server.

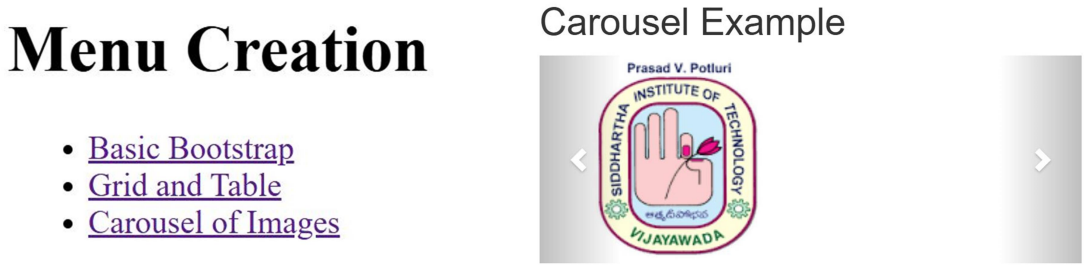
Finally Index. html

```
<!DOCTYPE html>
<html >
  <head>
      <title>Working with DJANGO - Creating Index</title>
  </head>
  <body>
    <h1>Menu Creation</h1>
    <ul>
      <li> <a href="/basicBS">Basic Bootstrap</a> </li>
      <li> <a href="/Grid_Table">Grid and Table</a> </li>
      <li> <a href="/Carousel">Carousel of Images</a> </li>
    </ul>
  </body>
</html>
```

views.py file

```
def index(request):
        return render(request, 'index.html')
```

urls.py file - - add - > path('',views.index),



Suresh Yadlapati